

Machine Learning-Based Auto Insurance Fraud Detection

Literature Review and Exploratory Data Analysis

Major Research Project

Master of Science in Data Science and Analytics

Name: Meshwa Patel

Student Number: 501390663

Supervisor: Shengkun Xie

Toronto Metropolitan University

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Abstract

Auto insurance fraud causes large financial losses for insurers and pushes up premiums for honest customers. This project focuses on the development of a supervised machine learning pipeline for identifying fraudulent auto insurance claims given the structure policy and incident data of a policyholder, such that the decision made can be explainable enough to aid in human investigator decision making. Given a dataset of 30,000 claims that has a known fraud flag, the performance of Logistic Regression, Random Forest and XGBoost models are compared. Each model is tested both with and without SMOTE (Synthetic Minority Over-sampling Technique), and the models are evaluated based on Precision, Recall and F1-score and AUC-ROC instead of accuracy due to a severe class imbalance in the dataset. Exploratory analysis reveals that the fraud signal present within the data is distributed lightly across multiple features. Incident severity is the most indicative feature. Using SMOTE, Logistic Regression obtains validation F1 of 0.310 and serves the purpose of a triage tool to rank claims according to the risk, rather than automatic classification. SHAP values were applied for both global and per-instance feature explanations, and a transparent rule based Fraud Risk Score was generated for the use of insurance investigator. This document describes the updated abstract, the literature review, data description and exploratory data analysis, and approach for the project.

1. Literature Review

1.1 Introduction

Insurance fraud is an old practice that has always existed. Drivers arrange to have accidents and sometimes make false statements about an accident and have additional items in an insurance claim that have not been damaged or are completely new, or can even manufacture an entirely fictional claim, resulting in money being paid by the insurer for something they never have been liable for, and the premium is passed back through to the customer for whom fraud costs every one of us as we subsidize a minority. Of all insurance types, motor insurance suffers because the number of claims is so great, while, at the same time, not all individual claims will warrant detailed manual investigation.

For many years the response was manually reviewing claims and applying pre-set rules. An experienced claim adjuster might just have a "feeling" that something wasn't quite right about the file, or a rule might flag every claim that had a value of above a certain dollar amount. Both of those approaches have their merit, but neither is a solution for the sheer volume of claims being submitted daily: "human review becomes a bottleneck, and even fairly sophisticated rules are easy to learn and navigate around for a sufficiently motivated fraudster." This is what machine learning aims to solve: instead of hardcoding rules into the system, an algorithm is "trained to recognize the pattern of past fraudulent and non-fraudulent claims" and then uses that knowledge to assign a "score to incoming claims based on the likelihood of them being fraudulent".

Here I outline what has been done already in the area and (crucially) what space exists to be explored. I will progress through the difficulties that fraud inherently presents, outline the major types of ML approaches applied to the task, consider in more depth the treatments of class imbalance and explainability by existing researchers in the area, and then present the four relevant works that provided the most inspiration for my own project. I conclude the chapter by highlighting the space I identified and providing a comparison of approaches.

1.2 The Nature and Cost of Insurance Fraud

I will begin with an overview of the problem itself, before diving into algorithms. One of the first lucid overviews of insurance fraud as a business problem comes from Viaene and anproblemse (2004) who set out the ways fraud can occur and why controlling it is so difficult. Crucially, as Viaene and anproblemse explain, fraud exists along a spectrum from simple exaggeration all the way to organized criminality. This spectrum is still relevant to modelling as the two extremes often don't appear the same in the data.

Brinkmann (2005) frames the question differently-why do ostensibly honest customers lie to their insurers at all? This behavioral perspective helps to remind that fraud is not purely a problem of technical detection; it is people acting upon opportunities, rationalizing behaviors, and this is in large part why the signal within the data is so tenuous. Aslam et al. (2022) provides a contemporary view with a review on the use of artificial intelligence and machine learning in insurance fraud, asserting that the landscape has long since moved on from manual methods.

What I got from those is that fraud detection is not going to be a 'set-and-forget' push-button job any time soon. The behavior is hidden on purpose, it moves around, and the cost of failure cuts both ways - miss the fraud, and you lose money, stop the innocent, and you erode confidence. Any model has to cope with that, not eliminate it.

1.3 From Manual Review to Data-Driven Detection

The move from human judgment to data-driven detection didn't take place all at once. Ngai et al (2011) published what is probably the most-cited review of data mining for financial fraud, and their classification framework is still a useful way to get around. They catalog which techniques, ranging from logistic regression to decision trees and neural networks, have been applied to which types of fraud, and their review is one of the reasons I felt comfortable retaining logistic regression as a baseline instead of dismissing it as too trivial. Kemp (2010) offers a related practical insight from the public sector-that fraud departments already had trouble with volume and automated assistance was necessary.

But the more global story here is that the ML revolution wasn't contained. It seeped into the neighbouring risk problems of loan default (Liu et al., 2022) and solvency prediction (Abdullah, 2021) and tells you how methods are transferable across tasks with roughly similar forms-structured data;

a rare negative event; predict rank-ordering by risk. Fraud detection is just one of those neighbouring problems.

1.4 Machine Learning Approaches to Fraud Detection

Once one takes for granted that the problem is a supervised one, there remains the question of what methods to employ. The literature can be split fairly naturally into several groups and it's probably worth going through them since my own choices are best understood relative to them.

1.4.1 Supervised classification on insurance data

Of the papers I can identify whose nearest to my work, all use supervised classifiers on labeled insurance claims data. Severino and Peng (2021) fit several machine learning methods on actual property-insurance micro-data and found that tree ensembles performed reasonably well given their nature as structured tabular data. I found this sufficiently motivating as to include not just logistic regression, but a tree ensemble (Random Forest and XGBoost) as well and not wager all on one model. Owolabi et al. (2023) actually set out to accomplish a very similar thing by fitting multiple classification models on the auto insurance fraud dataset and comparing them; this reinforced the idea that a straight model-to-model comparison would not be seen merely as an undergraduate exercise.

1.4.2 Lessons borrowed from credit-card fraud

Card fraud is an older sibling; many useful pieces of machinery dealing with rare events were invented there first. Adewumi & Akinyelu (2017) review machine-learning and nature-inspired approaches in card fraud, and Carneiro et al. (2017) survey a live data mining system implemented in an actual e-commerce environment, which serves as a helpful shot in the arm regarding what is practical in real life, beyond a research notebook. Van Vlasselaer et al. (2015) introduced network-based features through their APATE system and Forough & Momtazi (2021) & Zhang et al. (2021) move into deep sequential and feature-engineering-heavy deep learning approaches. Singh & Jain (2021) are of interest for a smaller point: they combine a bio-inspired model and a correlation-based feature selection method, and this concept of feature-selection has fed directly into the way I decided to prune my feature set.

I haven't applied many of these heavier models, and frankly it's because they aren't a good fit for the problem as I've formulated it. It's true that deep and network models do well on very large transaction volumes with complex, relational data. This is a claim data set of few tens of thousands of records, all structured, which is perfectly in the domain where standard classifiers can compete, and are much easier to explain. Taking lessons from this literature on handling imbalance and feature-selection but setting aside the heavier deep-learning models seemed like a good compromise given the importance placed on interpretation.

1.4.3 Deep learning, text, and network methods in insurance

Within insurance specifically, a couple of researchers have gone further than pure tabular models. Wang and Xu (2018) used deep learning combined with topic-model text analysis to parse the free text linked to auto claims - this is conceptually neat, because the free-text narrative describing the accident will likely have signal not contained within the structured fields. Van Vlasselaer et al. (2017), on social-security fraud detection with their work on GOTCHA, illustrate how powerful network analysis can be when you can chain together claimants; while interesting, these works require data that is completely unavailable to me – claims narrative or a relationship network of policyholders. While there is a justifiable limit of my project, it would be the obvious thing to investigate if I had such data.

1.4.4 Unsupervised and anomaly-based methods

Not all of the insurers are there with clean fraud labels to study, there is a line of research devoted to this kind of situation. Nian et al. (2016) discover the auto insurance fraud using an unsupervised spectral ranking method for anomalies, Gomes, Jin and Yang (2021) uses unsupervised deep learning on the same type of global issue. My project is supervised, as my major dataset does come with fraud label, so I do not pursue in this direction. But it is the most appropriate tool in the case of label scarcity or low trust of label, it shows my supervised approach is a bit luxury by relying on this conveniently labelled dataset.

1.5 The Class Imbalance Problem

If there is a single problem that all fraud papers have to face, it is class imbalance. True transactions are orders of magnitude more frequent than fraud, often by a factor of ten or more. What is awkward about this is that a model can achieve over ninety percent accuracy by always predicting fraud never happened, while flagging none. Accuracy can tell you almost nothing, so papers instead list precision, recall, F1, and AUC.

The standard way of addressing this imbalance is by resampling the training set and the most popular tool is SMOTE, which works by creating synthetic fraudulent data points so that the classifier is actually exposed to the minority class long enough to pick out some information from it. Komsrimorakot and Siriborvornratanakul (2025) investigate this rigorously in a motor insurance context, by presenting a combination of Center Point SMOTE and undersampling which was then validated on three public data sets. Their hybrid resulted in significantly more recall of the fraudulent class consistently across all data sets without degrading precision too much. I think I should flag the opposing argument, though. Ekici (2025) showed that SMOTE and class weighting does not work for all models or all data and can decrease performance. My results fall somewhere in between: SMOTE drastically increased recall while almost leaving the AUC untouched. My primary contention is precisely the reason for my experimentation – there should be no reporting of numbers from class-

weighted or SMOTE-treated datasets without at least a comparable number on the raw dataset, yet this is the case in many instances.

1.6 Making Models Explainable

A fraud model cannot be the only authority on a claim. A claim that the model flags has to be seen by an investigator who must determine whether to pursue the claim, request further documentation, or simply pay out the claim. A naked number that the investigator doesn't know the story behind is of little use, and rejecting claims based on a black box is asking for problems of fairness and legibility. Ekici (2025) presents an application of this on auto insurance fraud models using SHAP to obtain a ranked list of features which led to a particular prediction, and Carmona et al. (2022) present a similar application on predicting business failure using an XGBoost classifier whose title literally and figuratively proclaims that the black box era is finished.

Reading Ekici was the point where my own plan came into focus. It's true that SHAP gives good insight into a model, but its output is still a technical graph and aimed at a ML-savvy user. The end-user, the investigator, seldom receives an interface which they can easily read at their own pace. That realization is what drove me towards implementing a simple rule based risk score on top of the model, which I describe in the gap analysis section.

1.7 The Studies Closest to My Project

There are four new papers nearby enough that the project is worth a deeper dive. Three are mapping nicely to the three jobs the project tries to do all at the same time: Compare apples-to-apples. Do an honest job with the imbalance. Explain the result.

1.7.1 Ding et al. (2025): tuning XGBoost with particle swarm optimization

Ding, Ruan, Wang & Liu (2025) also work on auto insurance fraud using gradient boosting. The core strategy adopted by these researchers is the integration of XGBoost with a particle swarm optimization algorithm, where the optimal hyperparameters of the XGBoost algorithm were determined through automated searching, not by trial and error methods. They presented performance metrics worth reporting since these highlight the importance of model selection. While their model achieved almost 95%, (approximately 3% better than a basic XGBoost), standard approaches like SVM, Naive Bayes, logistic regression, and a back-propagation neural network resulted in 54.1%-68.6% accuracy, and Random Forest performed close to 78.4%. To explain how the model was working, SHAP was utilized to show that the most influential factors driving fraudulent claims were claim amount and vehicle type.

I read this paper partly as a warning: they extract every possible thing from one complex model but do so with an added level of complexity and computation which risks making the result both more difficult to reproduce and to explain. Although 95% is high, it's highly tuned on their dataset, and I

strongly suspect a number like that would not survive exposure to a more messy, low-signal dataset such as my own. My project makes the opposite move of trying to compare straightforward, fair comparisons between more or less standard models, which is a different task, rather than a less successful task.

1.7.2 Ekici (2025): interpretability as the main event

In their master's thesis at Tilburg University Ekici (2025) views interpretability as the objective rather than something to add afterwards. The SHAP model is used for feature importance and is also taken to the next level for local interpretability by using waterfall plots demonstrating how each individual feature pushes a single claim towards a fraud prediction or away. The master's thesis continues even further for logistic regression and provides results by coupling the SHAP model to standard statistical inference by stating coefficients and their p-values. The results obtained from the SHAP model show good consistency across all of the models where a couple of features (namely base policy type, vehicle price and month/year of the claim) end up in the top fifteen for every model.

Two parts of Ekici's work directly informed what I do here. First, the waterfall idea is precisely the per-claim explanation I wished to provide for an investigator, and I plot my data in the same way. Second, even more remarkably, Ekici shows that imbalance-handling methods such as class weighting and SMOTE can have varying, even negative, effects on a model depending on what measure you are optimizing against. This matches my findings where SMOTE sharply increased recall, while doing virtually nothing to AUC, so it is somewhat disheartening, though verifying, to find another published thesis reporting such results. While Ekici stops at the SHAP plot, my intent here is to turn the signals from that plot into an investigator's ready, out-of-the-box score.

1.7.3 Komsrimorakot and Siriborvornratanakul (2025): imbalance done carefully

This paper in Journal of Big Data is arguably the one that takes class imbalance most seriously. It eschews using just SMOTE, proposes a combined strategy, Center Point SMOTE (CP-SMOTE) plus random under-sampling, and tests it thoroughly against numerous other over-, under-, and hybrid techniques using three publicly available motor-insurance datasets that had various class imbalances. The headline is that CP-SMOTE combined with under-sampling is shown to consistently improve recall on the fraud class with a reasonable level of precision and AUC PR. The novelty is that this has been validated against three datasets, not one. This is precisely why I always train two of each model, one on original imbalanced data and one after oversampling with SMOTE, to highlight what resampling has achieved rather than taken it as granted.

1.7.4 Dondapati (2025): a recent comparative overview

The National High School Journal of Science has a 2025 publication by Dondapati that summarizes and tests multiple complex models on auto insurance claim fraud, evaluating how they perform

relative to each other. This paper is a good up-to-date overview of possible model options that is easily digestible and corroborates the importance of comparing classifiers as opposed to picking one that seems appropriate. I aim for a similar comparative intent, but under the constraints of identical settings on one data set, and I add layers of explainability, and risk-scoring not specifically explored in this paper. I want to also point out that this citation was attributed to the wrong person in my proposal.

1.8 Gaps in Literature

From looking through all this, I found four gaps which all occurred and each of these then became a design choice on my project and so have been displayed like so.

Gap 1 — Fair, like-for-like model comparison is rarer than you would expect.

There are many papers where only a specific state of the art model is tuned (Ding et al.) or where each model is individually optimized and pre-processed, creating bias. A clean study can be done with the same simple models, same data, and same preprocessing. That is exactly what I have done with Logistic Regression, Random Forest, and XGBoost.

Gap 2 — The effect of SMOTE is usually asserted, not shown.

Resampling is found everywhere but lack is the before/after comparison. I run all the models with SMOTE and without SMOTE, report both. In this way, you can see for yourself the actual cost/benefit of using resampling.

Gap 3 — Explainability tends to stop at the data scientist.

An investigation using a model such as that described by Ekici (2025) will make it clear to a technical audience how the model works, however it rarely results in an instrument the investigator can use effectively. I will attempt to bridge this by converting the model signals into an obvious rule based Fraud Risk Score and with clearly defined point values and Low, Medium and High bands.

Gap 4 — Honest reporting on hard, weak-signal data is uncommon.

Several papers report results with accuracy that is often not reproducible when run on real-world dirty data. I elected to submit modest but realistic results on a hard dataset and run the same method again on out-of-sample data which I believe is more illustrative of actual model performance.

1.9 What Others Did vs. What I Am Doing

The table below pulls the comparison together, so the contribution of my project is concrete rather than implied.

Aspect	What previous studies did	What I do in this project
Models	Generally, a single tuned model (e.g. PSO-XGBoost in Ding et al.) or models trained for different terms.	Logistic Regression, Random Forest and XGBoost run under identical preprocessing for a genuinely fair comparison.
Class imbalance	SMOTE or under-sampling applied, with only the rebalanced result reported (e.g. Komsrimorakot & Siriborvornratanakul).	Every model run with and without SMOTE, so the real effect of resampling is measured, not assumed.
Explainability	SHAP used to explain models to a technical audience (e.g. Ekici; Carmona et al.).	SHAP plus a plain rule-based Fraud Risk Score (Low/Medium/High) built for non-technical investigators.
Evaluation	Sometimes, I have high accuracy on clean or favorable datasets.	Honest, modest results on a hard dataset, then re-tested on independent datasets to show how performance shifts.
Output	A trained model and a table of metrics.	A trained model, a per-claim explanation layer, and a usable risk-scoring tool an investigator can read.

1.10 Conclusion

From all this, it is abundantly clear that ML is the new status quo for auto insurance fraud, and the same four problems (class imbalance, faint/obscured signal, and humanly understandable decisions) will persist over time. Work in the past couple of years has continued to make progress in all four areas, from Ding et al. (2025) and Komsrimorakot and Siriborvornratanakul (2025) through automated tuning of ML parameters and detailed analysis of imbalance respectively, to Ekici (2025) providing SHAP-based model interpretability; at the same time, foundations in prior work such as Ngai et al. (2011), Nian et al. (2016), Severino and Peng (2021), and across credit-card fraud literature generally remain dominant for methods.

What I did not find was a study which would do all the unglamorous things really well together: give the standard methods a fair comparison, quantify rather than hypothesis the effect of SMOTE, pass the interpretation to the human researcher rather than to the data scientist, and reveal what concrete data supports the finding, unembellished. It is the conjunction of these practices that my MRP will try to fill, it will not produce the highest value of accuracy in literature and, given the data involved,

should not pretend to. It is the careful, interpretable and genuinely assessed pipeline that an insurance firm would trust.

2. Data Description and Exploratory Data Analysis

2.1 Dataset Summary

The dataset is the open-source Car Insurance Fraud Dataset which consist of records for U.S auto insurance claims downloaded from Kaggle. This is an open-source dataset where each record has a confirmed binary flag indicating the claim is fraudulent or not which makes it perfect for supervised classification. It consists of 30,000 individual auto insurance claim records ranging from November 2023 to November 2025 with 24 features along with a target variable per claim. The features are divided naturally into customer characteristics (age, sex, education, occupation, hobbies), policy characteristics (state, deductible, annual premium) and incident characteristics (incident type, collision type, severity, whether authorities were involved and if witnesses were called, claim amounts, and location). The target variable for classification is `fraud_reported` where Y represents fraudulent claims and N non-fraudulent claims.

2.2 Descriptive Statistics

The data set is clean and well organized. The data set is highly imbalanced. The major descriptive statistics of the data set are given as below.

Statistic	Value
Total claims	30,000
Fraudulent claims	3,440 (11.47%)
Legitimate claims	26,560 (88.53%)
Class imbalance ratio	approximately 7.7 to 1
Number of features	24 (plus the target)
Duplicate records	None
Columns with missing values	1 (<code>authorities_contacted</code> , 25.2% missing)
Date range	9 November 2023 to 8 November 2025
Mean age (fraud vs non-fraud)	44.5 vs 46.8 years

2.3 Constraints and Preprocessing

There were two constraints during the analysis. First, there is such a high degree of class imbalance (only 11.5% fraud) that accuracy is an unsuitable metric and resampling must be employed in modeling. Second, exploratory analysis indicated that the predictive signal is weak and diffuse across the many features rather than strong in any given feature, which places a ceiling on what any model is capable of on this dataset. The cleaning was purposefully minimal as the data was already

relatively clean. The cleaning performed were fill blank authorities_contacted with 'None' (reasoning that a blank meant that no authority had been contacted); convert incident_date to date type; trim white space from text categories; convert fraud_reported to numeric Y/N values to 1/0; high cardinality categorical columns policy_id and incident_city were dropped from modeling but retained for context.

2.4 Exploratory Data Analysis

There was really only one result of this exploratory analysis and that is that the fraud signal in the data is faint and diffuse. As can be seen from Figure 1 the target has class imbalance and is also by far the most significant attribute in data, which will guide the modeling choices.

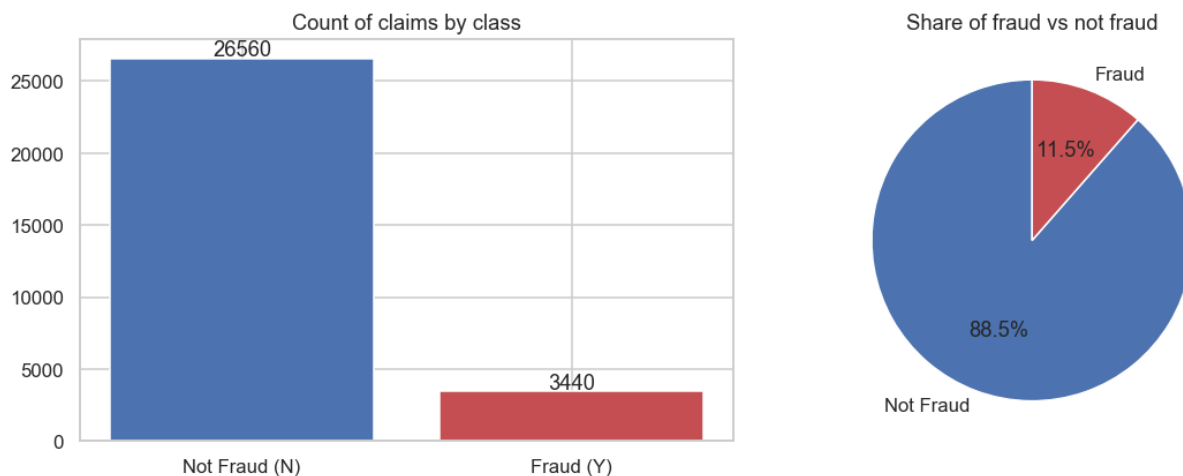


Figure 1. Distribution of the target variable (fraud vs. legitimate).

When comparing demographics of claimants (Figure 2) they showed little correlation with fraud, Sales and Clerk jobs showed a slightly increased fraud rate and Teacher showed a slightly decreased rate, fraudulent claimants were on average slightly younger but none of these impacts is individually significant.

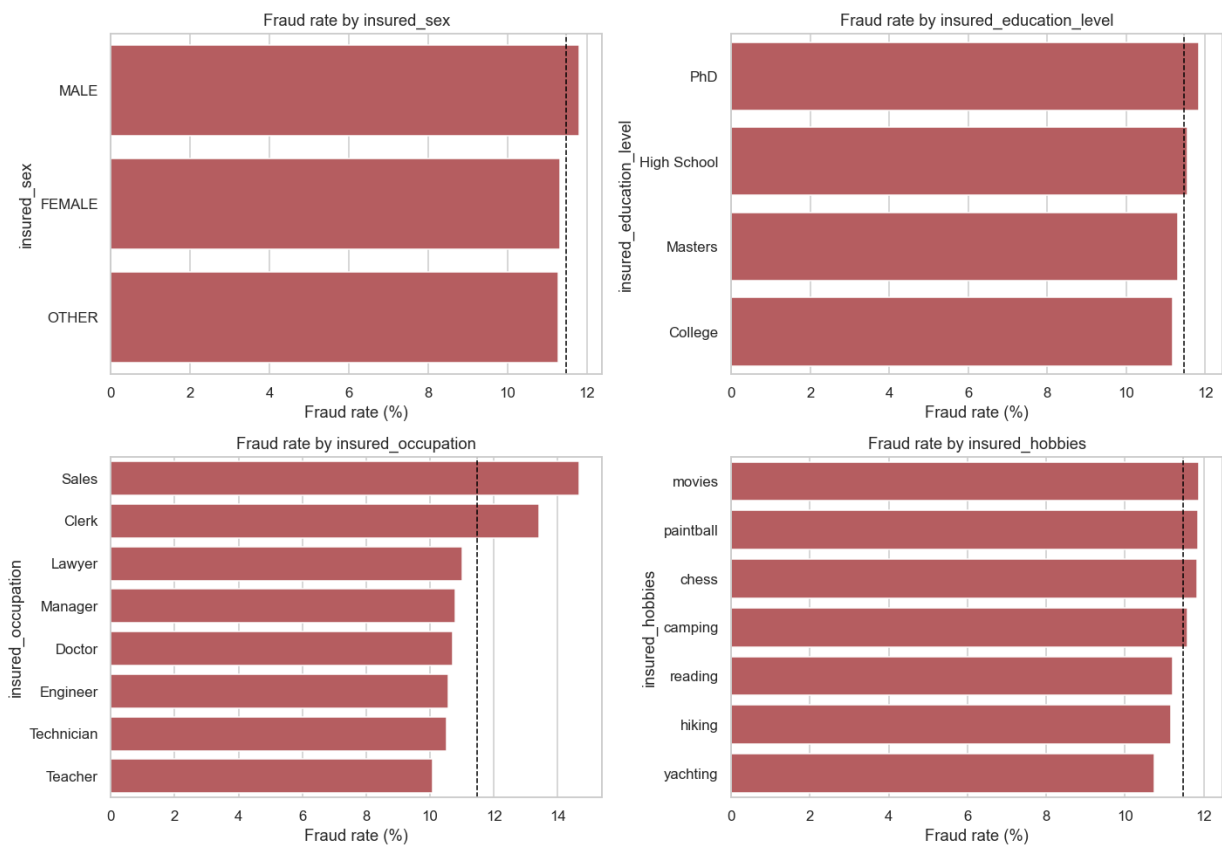


Figure 2. Fraud rate by customer demographic features.

The best single indicator found in the data was with regards to severity of loss (Figure 3). "Total Loss" incidents had a fraud percentage around 14.8% while "Minor Damage" was around 9.6%. This was the single best predictor.

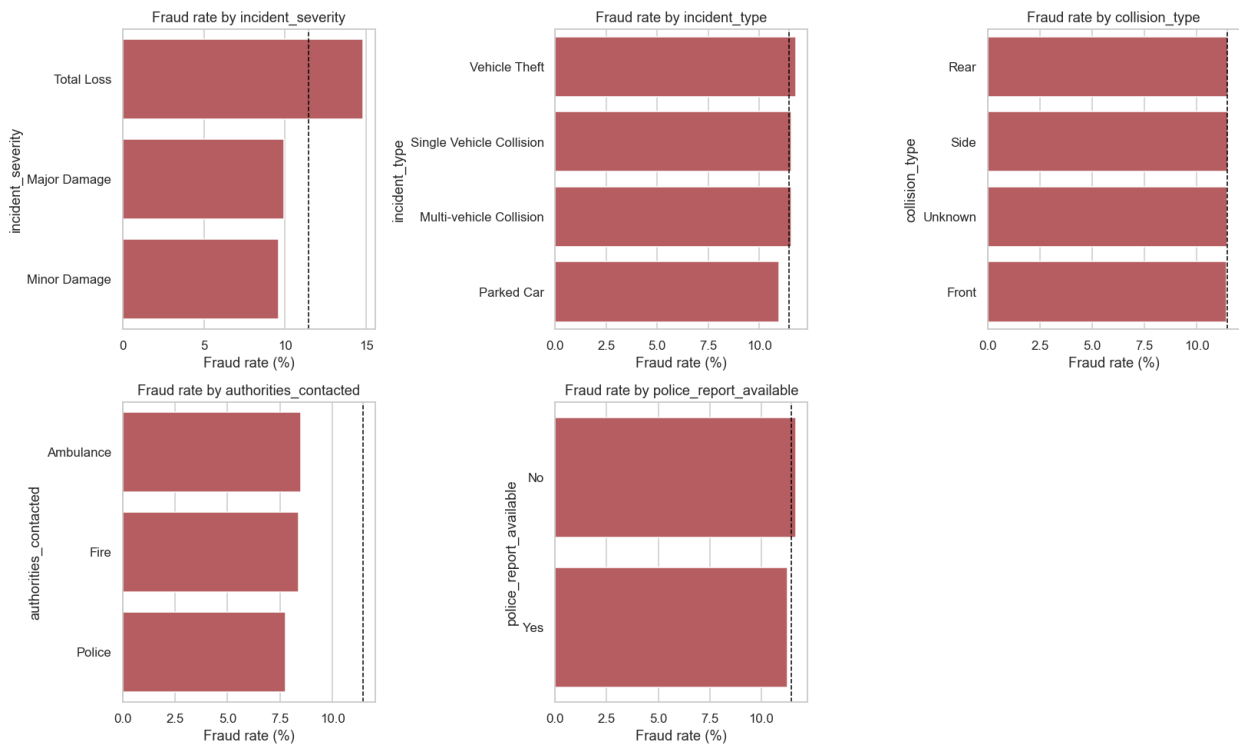


Figure 3. Fraud rate by incident features; incident severity shows the clearest separation.

Fraudulent claims were related to somewhat larger amounts of claims (Fig 4) however distribution is largely overlapping again which makes the signal a weak one.

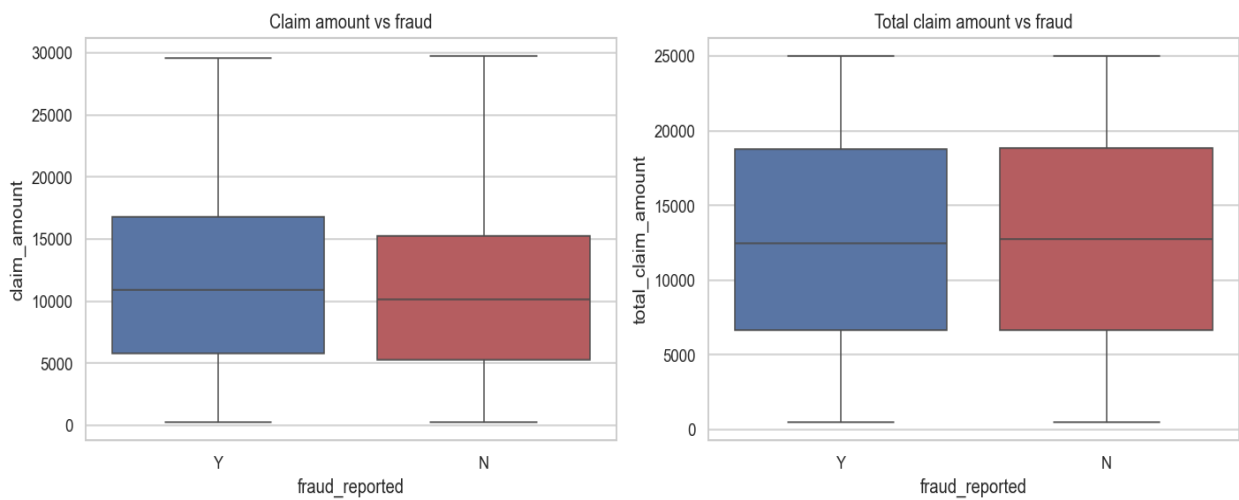


Figure 4. Claim amount distribution by fraud status.

The correlation heatmap of the numeric features also painted a similar picture (Figure 5): None of the features are significantly correlated to the target (fraud) and no two features have strongly correlated against one another. This makes sense if the dataset is designed such that the evidence for fraud has to be deduced from multiple subtle clues rather than one dominant factor.

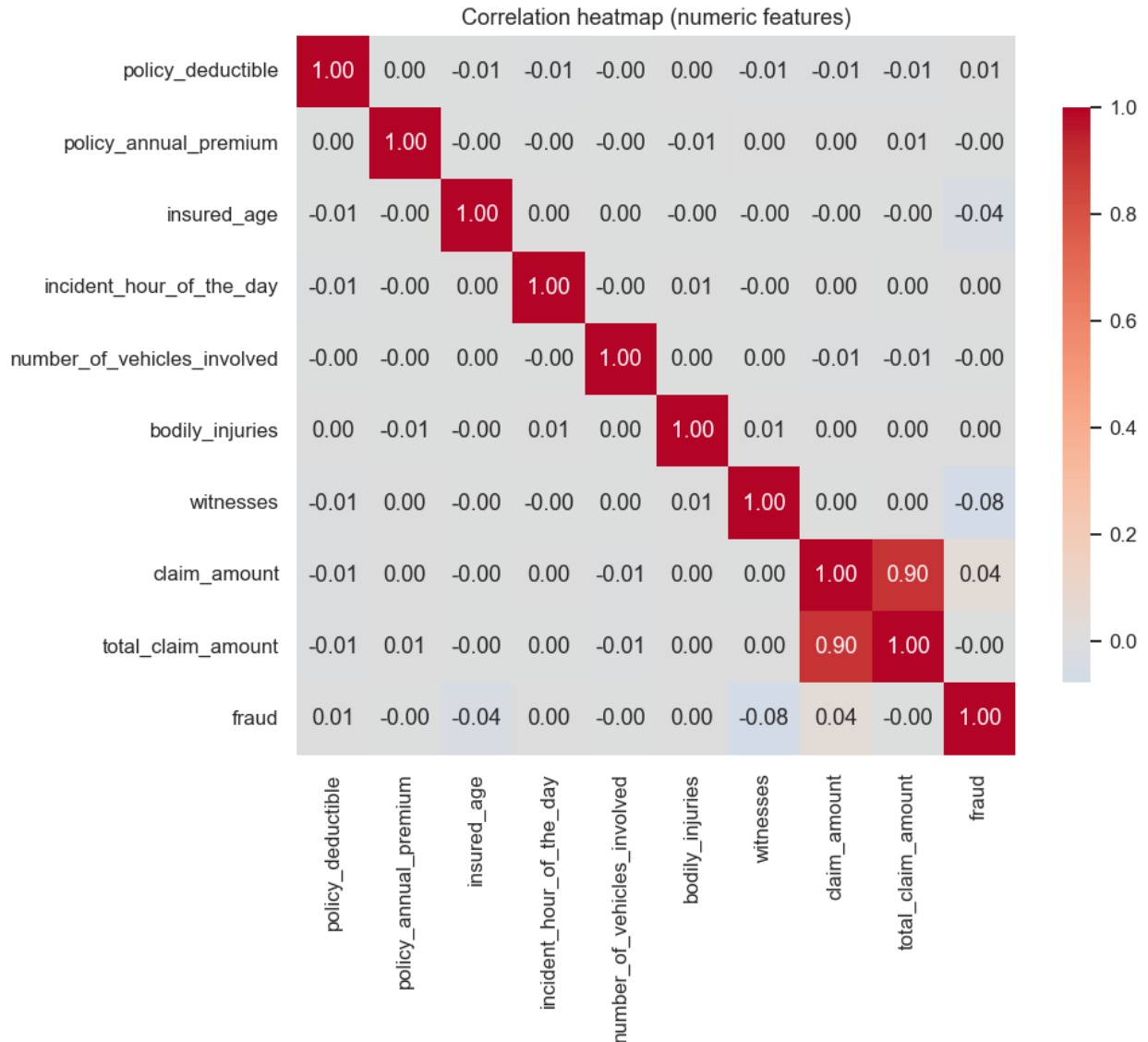


Figure 5. Correlation heatmap of numeric features.

The temporal and spatial trends in the fraud rates were also analyzed. Figure 6 is the time trend of fraud per month, and no seasonal trend appears. Figure 7 is the state level fraud per month. There appears to be little difference between the ten states, except maybe slightly more Fraud in a couple of states. So, it appears to be reasonable that there is no trend in space or time, given that there is no trend at all.

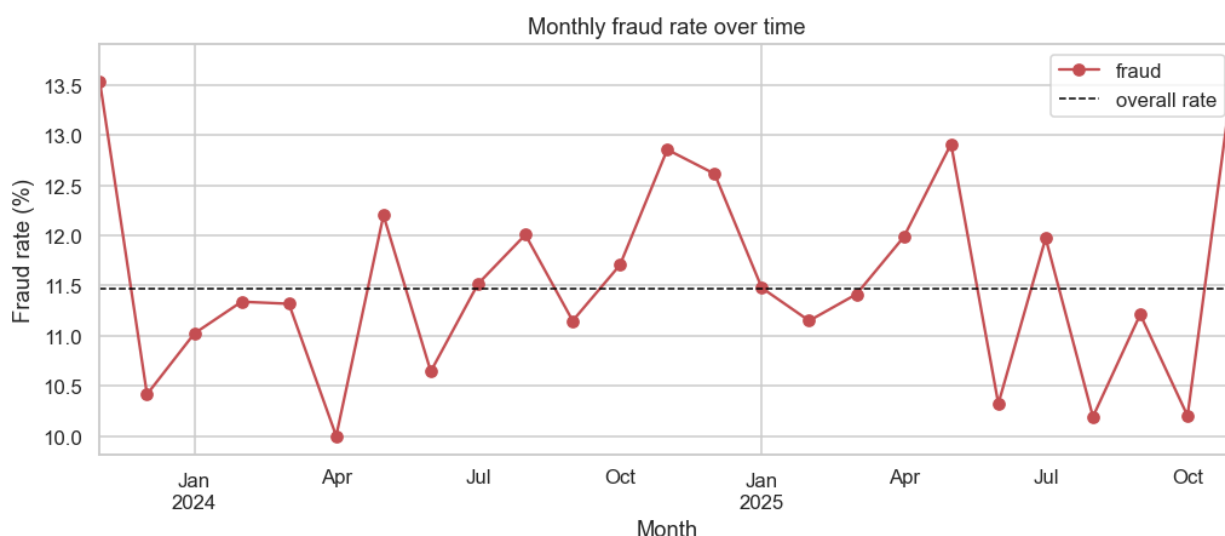


Figure 6. Monthly fraud trend over the period covered by the dataset.

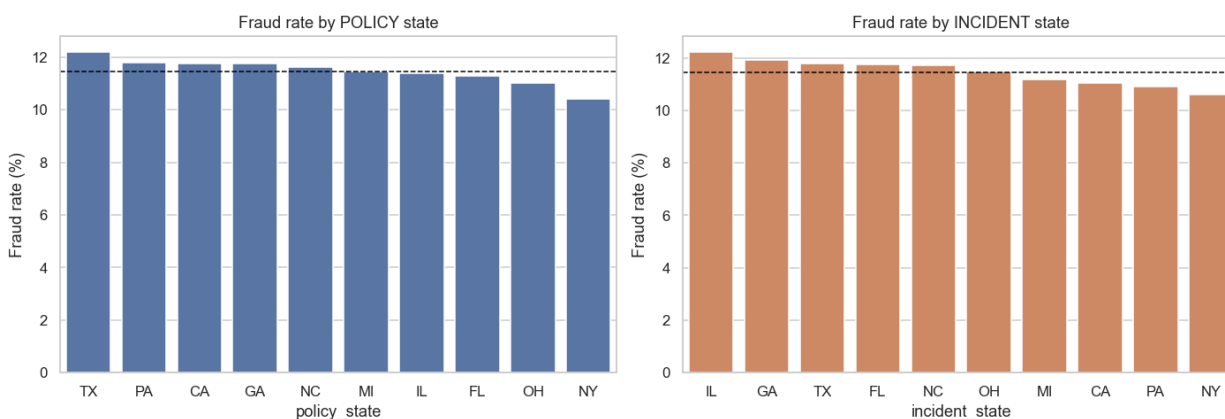


Figure 7. Fraud rate by state.

3. Project Approach

3.1 Research Methodology

The approach follows the typical workflow of supervised machine learning, slightly adapted to account for the two constraints established previously (class imbalance, weak signal). The data, after being cleaned, goes through exploratory analysis, feature engineering and then a stratified 70/15/15 train, validation, and test split ensuring that fraud rate is consistent across subsets. Numeric are standardized and categorical variables are one-hot encoded via a single column transformer to apply the same preprocessing pipeline across all splits and avoid data leakage. Three classifiers-Logistic Regression, Random Forest and XGBoost-are each trained on both the raw imbalanced data and also with SMOTE applied within the pipeline (total 6 variations), and assessed with precision, recall, F1-score, and AUC-ROC. The winner based on validation F1 is then explained

with SHAP and integrated into a transparent, rule based Fraud Risk Score. A check on fairness and assessment on external datasets close the pipeline.

3.2 Methodology Diagram

Figure 8 illustrates the overall project methodology from raw data through to external validation.

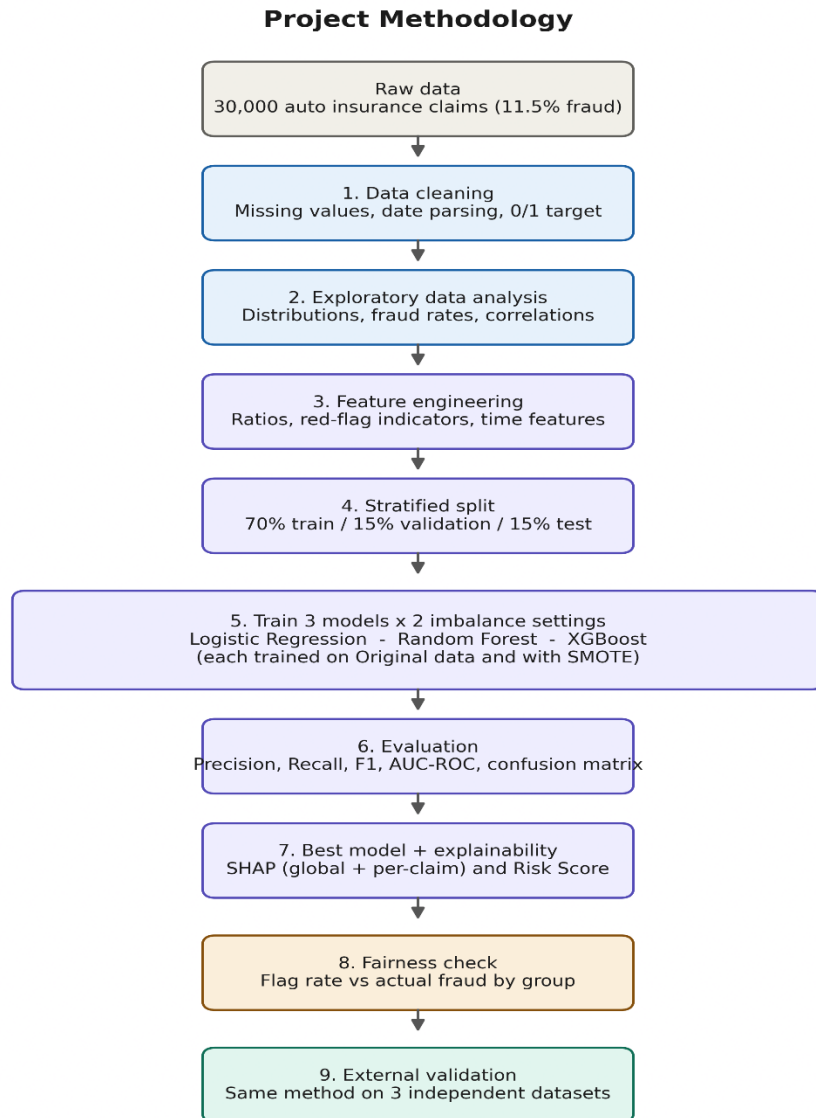


Figure 8. Overall project methodology.

3.3 Justification of the Approach

Each choice is grounded in the literature reviewed in Section 1. The three models were chosen to span a simple, transparent baseline (Logistic Regression) and stronger ensemble methods (Random Forest and XGBoost) that Severino and Peng (2021) and Ding et al. (2025) found effective on structured insurance data. Running each model with and without SMOTE directly addresses the gap that resampling effects are usually asserted rather than measured and reflects the cautionary finding of Ekici (2025) and Komsrimorakot and Siriborvornratanakul (2025) that imbalance handling does not always help. Selecting by F1 rather than accuracy follows directly from the class-imbalance literature, since accuracy rewards a model that never predicts fraud. The use of SHAP follows Ekici (2025) and Carmona et al. (2022) and extending it into a plain-language risk score addresses the gap that explainability rarely reaches the investigator. Finally, validating the same pipeline on independent datasets provides an honest check on whether the method generalizes, rather than relying on a single headline number.

4. GitHub Repository

The complete project, including the four Jupyter notebooks (data cleaning, exploratory data analysis, modelling, and external testing), the cleaned dataset, all figures, and supporting documentation, is available at the repository below.

Repository link: <https://github.com/MeshwaPatel-11/MRP-Auto-Insurance-Fraud-Detection>

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